



**DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING**

## ***Experiment No.: 4***

### ***Interfacing LPC2148 for internal ADC and program to display digital value on LCD***

Name: \_\_\_\_\_

Roll No.: \_\_\_\_\_ Batch: \_\_\_\_\_

Date of Performance: \_\_\_\_\_

Date of Assessment: \_\_\_\_\_

| Particulars                 | Marks |
|-----------------------------|-------|
| Marks for Regularity (05)   |       |
| Marks for Performance (15)  |       |
| Understanding (Viva) (05)   |       |
| Total (25)                  |       |
| Signature of Course Teacher |       |

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**Experiment: 4**

**AIM:** Interfacing LPC2148 for internal ADC and program to display digital value on LCD.

**OBJECTIVE:** To understand the working of the on-chip Analog-to-Digital Converter (ADC) in LPC2148.

**APPARATUS:**

1. LPC2148 module development board.
2. Connector wire
3. Adapter

**THEORY:**

Most of the devices with which we interact will be digital, or consist of only two possible 'states' (On/Off, True/False, etc.). Almost every dial or knob on any modern electronics device, for example, is probably analog. Since our microcontrollers are digital, what that means is that we need to find a way to convert those analog signals into something 'digital' that our microcontroller can actually understand. That's where Analog to Digital Converters (ADCs) come in, and thankfully for us the LPC2148 has two of them built in. ADCs essentially act as a bridge (or a 'translator') between the messy outside analog world and the cozy, black and white (green and white?) digital world our microcontrollers live in. They work by converting voltage to a numeric value that the microcontroller can understand.

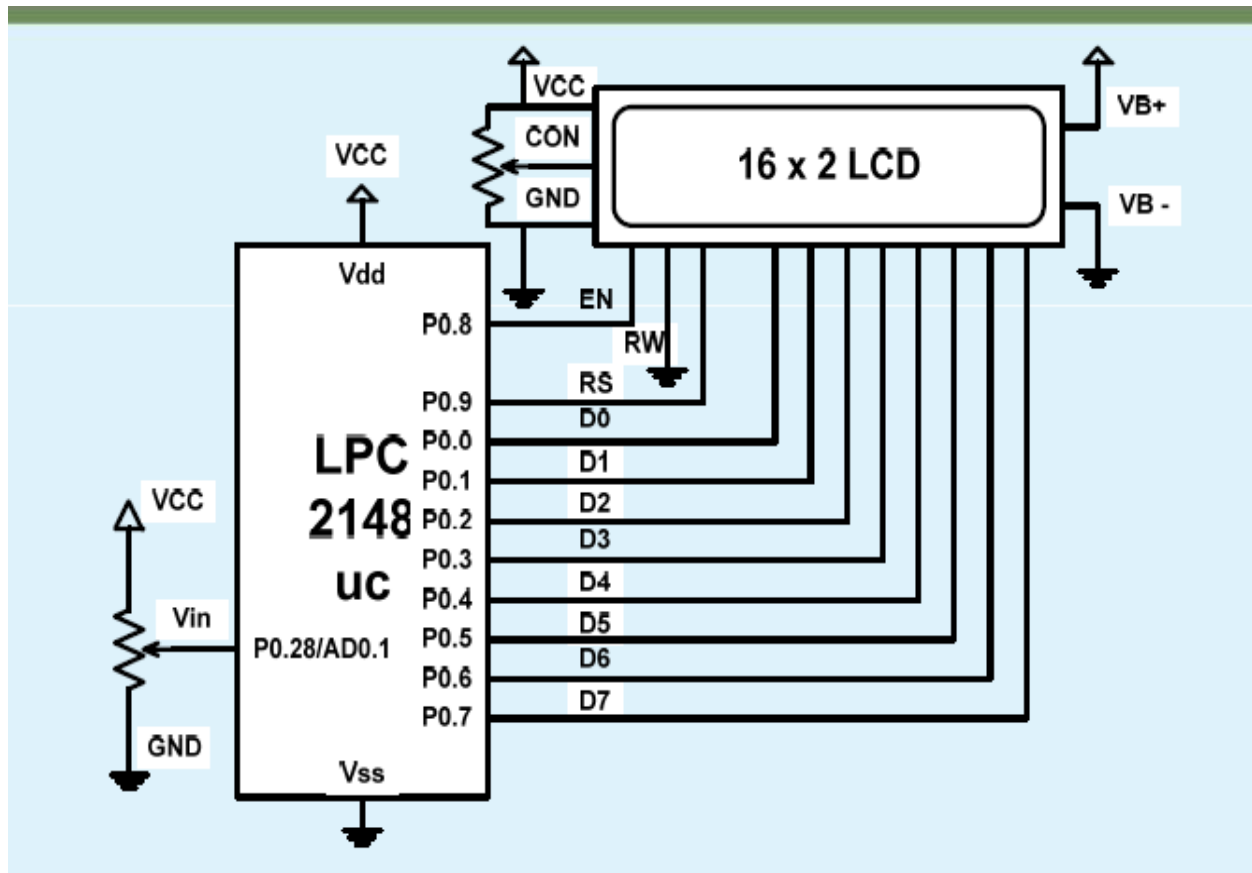
For example, with an internal voltage of 3.3V (which is the  $V_{ref}$  on the LPC2148) and your ADC set to return the maximum 10-bit data (meaning you have possible values between 0 and 1023), 0.0V would return 0, 3.3V (or higher) would return 1023, and 1.65V would return ~512.

They only work in one direction (reading data from outside and sending it 'into' the chip), but life without them would be a lot more challenging ... or at the very least a lot more expensive (analog devices are often much cheaper than their digital counterparts).

Basic clocking for the A/D converters is provided by the APB clock. A programmable divider is included in each converter, to scale this clock to the 4.5 MHz (max) clock needed by the successive approximation process. A fully accurate conversion requires 11 of these clocks.

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**INTERFACING DIAGRAM:**



**ADC FEATURES:**

- 10-bit successive approximation analog to digital converter (one in LPC2141/2 and two in LPC2144/6/8).
- Input multiplexing among 6 or 8 pins (ADC0 and ADC1).
- Power-down mode.
- Measurement range 0 V to VREF (typically 3 V; not to exceed VDDA voltage level).
- 10-bit conversion time > 2.44  $\mu$ s.
- Burst conversion mode for single or multiple inputs.
- Optional conversion on transition on input pin or Timer Match signal.
- Global Start command for both converters (LPC2144/6/8 only).

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**REGISTERS:**

| ADC SIGNALS | PIN NO. | Symbol                    | DESCRIPTION                                                                                                                                                                                                                                                                                           |
|-------------|---------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AD0.1       | 13      | P0.28/AD0.1/CAP0.2/MAT0.2 | Analog Inputs. The ADC cell can measure the voltage on any of these input signals. Note that these analog inputs are always connected to their pins, even if the Pin Function Select reg. assigns them to port pins. A simple self-test of the ADC can be done by driving these pins as port outputs. |
| AD0.2       | 14      | P0.29/AD0.2/CAP0.3/MAT0.3 |                                                                                                                                                                                                                                                                                                       |
| AD0.3       | 15      | P0.30/AD0.3/EINT3/CAP0.0  |                                                                                                                                                                                                                                                                                                       |
| AD0.4       | 9       | P0.25/AD0.4/Aout          |                                                                                                                                                                                                                                                                                                       |
| AD0.6       | 27      | P0.4/SCK0/CAP0.1/AD0.6    |                                                                                                                                                                                                                                                                                                       |
| AD0.7       | 29      | P0.5/MISO0/MAT0.1/AD0.7   |                                                                                                                                                                                                                                                                                                       |
| AD1.0       | 30      | P0.6/MOSI0/CAP0.2/AD1.0   |                                                                                                                                                                                                                                                                                                       |
| AD1.1       | 33      | P0.8/TXD1/PWM4/AD1.1      |                                                                                                                                                                                                                                                                                                       |
| AD1.2       | 35      | P0.10/RTS1/CAP1.0/AD1.2   |                                                                                                                                                                                                                                                                                                       |
| AD1.3       | 38      | P0.12/DSR1/MAT1.0/AD1.3   |                                                                                                                                                                                                                                                                                                       |
| AD1.4       | 39      | P0.13/DTR1/MAT1.1/AD1.4   |                                                                                                                                                                                                                                                                                                       |
| AD1.5       | 45      | P0.15/RI1/EINT2/AD1.5     |                                                                                                                                                                                                                                                                                                       |
| AD1.6       | 1       | P0.21/PWM5/AD1.6/CAP1.3   |                                                                                                                                                                                                                                                                                                       |
| AD1.7       | 2       | P0.22/AD1.7/CAP0.0/MAT0.0 |                                                                                                                                                                                                                                                                                                       |
| VREF        | 63      | VREF                      | Provides a voltage reference level for ADC.                                                                                                                                                                                                                                                           |
| VDDA        | 7       | VDDA                      | Analog Power and Ground. These should be nominally the same voltages as VDD & VSS, but should be isolated to minimize noise & error.                                                                                                                                                                  |
| VSSA        | 59      | VSSA                      |                                                                                                                                                                                                                                                                                                       |

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**A/D Control Register (AD0CR - 0xE003 4000 and AD1CR - 0xE006 0000)**

**Table 280: A/D Control Register (AD0CR - address 0xE003 4000 and AD1CR - address 0xE006 0000) bit description**

| Bit   | Symbol | Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reset value |
|-------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 7:0   | SEL    |       | Selects which of the AD0.7:0/AD1.7:0 pins is (are) to be sampled and converted. For AD0, bit 0 selects Pin AD0.0, and bit 7 selects pin AD0.7. In software-controlled mode, only one of these bits should be 1. In hardware scan mode, any value containing 1 to 8 ones. All zeroes is equivalent to 0x01.                                                                                                                                                  | 0x01        |
| 15:8  | CLKDIV |       | The APB clock (PCLK) is divided by (this value plus one) to produce the clock for the A/D converter, which should be less than or equal to 4.5 MHz. Typically, software should program the smallest value in this field that yields a clock of 4.5 MHz or slightly less, but in certain cases (such as a high-impedance analog source) a slower clock may be desirable.                                                                                     | 0           |
| 16    | BURST  | 1     | The AD converter does repeated conversions at the rate selected by the CLKS field, scanning (if necessary) through the pins selected by 1s in the SEL field. The first conversion after the start corresponds to the least-significant 1 in the SEL field, then higher numbered 1-bits (pins) if applicable. Repeated conversions can be terminated by clearing this bit, but the conversion that's in progress when this bit is cleared will be completed. | 0           |
|       |        | 0     | Conversions are software controlled and require 11 clocks.                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| 19:17 | CLKS   |       | This field selects the number of clocks used for each conversion in Burst mode, and the number of bits of accuracy of the result in the RESULT bits of ADDR, between 11 clocks (10 bits) and 4 clocks (3 bits).                                                                                                                                                                                                                                             | 000         |
|       |        | 000   | 11 clocks / 10 bits                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
|       |        | 001   | 10 clocks / 9bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 010   | 9 clocks / 8 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 011   | 8 clocks / 7 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 100   | 7 clocks / 6 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 101   | 6 clocks / 5 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 110   | 5 clocks / 4 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|       |        | 111   | 4 clocks / 3 bits                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| 20    | -      |       | Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.                                                                                                                                                                                                                                                                                                                                          | NA          |
| 21    | PDN    | 1     | The A/D converter is operational.                                                                                                                                                                                                                                                                                                                                                                                                                           | 0           |
|       |        | 0     | The A/D converter is in power-down mode.                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| 23:22 | -      |       | Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.                                                                                                                                                                                                                                                                                                                                          | NA          |

**CONCLUSION:**